

Learning Robust Stair-Climbing Skills for Open Duck Mini through Reward Shaping and Curriculum Learning

Motivation and Research Question (WHY?):

- Biped robots are attractive for deployment in human-designed environments because they can potentially traverse curbs, stairs, thresholds, and other elevation changes that are difficult for wheeled platforms. Even a single small step, however, is a challenging contact-rich control problem: the robot must coordinate multiple joints, achieve sufficient foot clearance, place the foot accurately, transfer its center of mass, and remain balanced through changing contacts. A controller that works on flat ground therefore does not automatically generalize stair climbing.
- We will use Open Duck Mini in MuJoCo to study a focused version of this problem: climbing one rigid step and stabilizing on the upper surface. The project is not only about obtaining a successful policy; it asks a measurable learning question: how do reward shaping and curriculum learning affect sample efficiency, final success rate, and robustness for biped stair climbing? Simulation-to-real transfer will be treated as an optional extension rather than a required outcome.

Conventional Methods and Why RL?:

- Conventional biped locomotion controllers commonly rely on predefined gait trajectories, inverse kinematics, finite-state machines, trajectory optimization, or model-predictive control. Their main advantages are interpretability and the ability to encode explicit balance, contact, and kinematic constraints. Their downsides are that motion phases and trajectories may require substantial manual design and retuning when step height, friction, contact timing, or robot dynamics change. Accurate model-based optimization can also become difficult when repeated contact transitions and model mismatch are important.
- Reinforcement learning (RL) offers a complementary approach: instead of prescribing a complete climbing motion, we specify observations, actions, and task objectives and let the policy discover the required joint coordination through interaction. The main challenge is that sparse success rewards make exploration difficult and poorly designed rewards can produce undesirable behaviors. This motivates our two central techniques: task-specific reward shaping to provide informative intermediate feedback, and curriculum learning to expose the robot to easier steps before harder ones.

Problem Statement (WHAT?):

- The Open Duck Mini model starts in front of a single rigid step with an initial height of approximately 2 cm. The robot must approach the step, place one foot on the upper surface, move its body forward, bring the second foot up, and remain upright for a short settling period. A trial is successful when both feet and the base reach the upper surface within a time limit while the torso remains sufficiently upright. An episode terminates early if the torso height falls below a threshold; base tilt becomes excessive, or the robot leaves the task area.
- The policy receives proprioceptive robot state together with compact task information about the step; camera perception is deliberately excluded from the initial study so that the project focuses on locomotion and contact control. After the fixed-height task is established, we will test several step heights and compare direct training at a target height with a curriculum that progressively increases the height after the policy reaches a success-rate threshold.

Reinforcement Learning Formulation (HOW?):

- State / observation space: projected gravity or base orientation; base linear and angular velocities; joint positions and velocities; previous actions; left/right foot-contact or foot-height signals; distance to the step edge; step height; and relative location of the desired landing region.

- Action space: continuous desired joint-position offsets for the actuated leg joints. A low-level PD controller converts these targets into torques, with action bounds around a nominal standing pose to keep exploration physically plausible.
- Reward: $r = w_p r_{\text{progress}} + w_f r_{\text{foot}} + w_u r_{\text{upright}} + w_s r_{\text{success}} - w_e r_{\text{effort}} - w_a r_{\text{smooth}}$. Progress rewards approaching or advancing through the climb; foot-placement rewards clearance and landing on the upper surface; upright and success terms discourage falling-forward exploits; effort and smoothness penalties discourage large or jerky commands; falling receives a terminal penalty.
- Reward ablation: compare a baseline reward containing progress/approach, uprightness, and fall terms with a shaped reward that additionally includes foot-placement and staged progress terms.
- Policy network: separate PPO actor and critic multilayer perceptrons, each with two hidden layers of 256 units and nonlinear activations. The actor parameterizes a Gaussian distribution over continuous actions; the critic estimates state value.

RL Algorithm, Experiments, and Expected Outcomes:

- We will primarily use Proximal Policy Optimization (PPO), an on-policy actor-critic algorithm well suited to continuous robot control and parallel simulation. Training will use multiple MuJoCo environments, randomized initial poses, and modest friction variation. The main experimental comparison is a 2 x 2 design: (i) baseline vs. shaped reward and (ii) direct target-height training vs. curriculum learning.
- Evaluation metrics will include climbing success rate, average episodic return, training steps required to reach a target success rate, time to complete the climb, fall rate, and control/energy effort. We expect reward shaping to improve learning speed and final fixed-height success by providing denser feedback for difficult intermediate phases. We expect curriculum learning to improve stability and performance on larger steps by allowing the policy to first learn basic foot lifting, balance, and weight transfer on easier versions of the task.
- If time permits, we will strengthen domain randomization over friction, mass, and step position and consider sim-to-real transfer as an optional extension.